Project: Retail Analytics & Customer Churn Data Warehouse Pipeline

1- Project Proposal:

Objective:

To design and build an end-to-end data pipeline and Medallion Data Warehouse that consolidates multi-source retail data, automates data ingestion and transformation with incremental updates, and delivers executive star-schema dashboards for the CEO, CFO, and COO to analyze business performance and mitigate customer churn.

Description:

This project covers the full lifecycle of data engineering and business intelligence. Data is ingested from multiple sources (online database APIs via Supabase and static CSV files) into Microsoft SQL Server. The warehouse implements a structured Medallion architecture: a staging layer for initial landing, a bronze layer for raw data preservation, a silver layer for cleaning, standardization, and incremental upserts/merges, and a gold layer modeling facts, dimensions, and analytical views. A local Python orchestration script automates the entire ingestion and stored procedure execution workflow with single-click execution and error handling. Finally, the modeled gold layer is loaded into Power BI via Import Mode, leveraging pre-aggregated SQL views and Power Query transformations to produce role-specific analytical dashboards.

Tools & Design:

- **Architecture:** Medallion Data Warehouse Architecture (Staging, Bronze, Silver, Gold)
- **Data Modeling:** Star Schema (Fact and Dimension tables), Pre-aggregated Gold Analytical Views
- **Ingestion & Incremental Strategy:** Local single-click execution pipeline, Incremental Upsert / Merge logic using CTEs, Indexes, and Stored Procedures with TRY...CATCH error handling
- **Reporting Design:** Role-based executive views tailored for CEO, CFO, COO, and Customer Retention / Churn analysis

Technologies to Use:

- **Storage & Relational Database:** Microsoft SQL Server
- **Source Database:** CSV Files
- **Programming & Automation:** Python
- **Data Modeling & Visualization:** Microsoft Power BI (Power Query, Star Schema Data Modeling, DAX, Import Mode)

Team Members & Responsibilities:

1. **Mostafa Mahmoud Abdelhamid**
    - **Responsibilities:** Setting up the GitHub architecture, managing data sources (CSV and Supabase), building the staging layer, and establishing the SQL database connections.
2. **Noureldien Mohamed Mahmoud**
    - **Responsibilities:** Developing the Bronze and Silver layers (handling raw data ingestion, standardization, cleaning, and incremental logic).
3. **Momen Shreif Mohamed**
    - **Responsibilities:** Designing the Gold layer (Star Schema), creating all analytical SQL views, and providing overall team organization and assistance.
4. **Ali Mohamed Ahmed**
    - **Responsibilities:** Building and encapsulating the entire automated data pipeline using Python, and developing the Power BI data model to ensure fast and seamless queries from the Gold layer.
5. **Mohamed Gehad Mohamed**
    - **Responsibilities:** Serving as the primary BI Developer and Dashboard Designer. Tasked with building the final dashboards based on stakeholder user stories, designing an interactive UI/UX (drill-downs, slicers, tooltips), and conducting final data validation to guarantee visual accuracy prior to delivery.

Week 1: Setup, Source Ingestion, Staging & Bronze Layer

- Initialize the GitHub repository and set up version control architecture (Mostafa Mahmoud Abdelhamid)
- Establish connections to the local CSV files and build the Staging layer architecture in Microsoft SQL Server (Mostafa Mahmoud Abdelhamid)
- Load the raw retail and churn data into the Staging layer (Mostafa Mahmoud Abdelhamid)
- Create Stored Procedures to migrate raw data from the Staging layer to the Bronze layer to preserve an exact historical copy (Mostafa Mahmoud Abdelhamid)
- Perform Exploratory Data Analysis (EDA) on the raw data to validate the proposed stakeholder business questions and ensure they are achievable (Momen Shreif Mohamed)

- Review the ingested data in the Bronze layer and define the cleaning and standardization rules required for the next layer (Noureldien Mohamed Mahmoud)

Deliverables:

- GitHub repository active with proper team access.
- Fully populated SQL Server Staging layer containing the raw CSV data.
- Operational Bronze layer preserving an exact copy of the raw data.
- Completed EDA report finalizing and validating the stakeholder questions.

Week 2: Silver & Gold Layer Transformations

- Design the Silver layer schema to hold cleaned, standardized, and typed data (Noureldien Mohamed Mahmoud)
- Develop the SQL cleaning logic using CTEs and implement incremental loads (Upserts/Merges) in the Silver layer (Noureldien Mohamed Mahmoud)
- Design the Gold layer Star Schema (Fact and Dimension tables) required for retail and churn analysis (Noureldien Mohamed Mahmoud)
- Develop Stored Procedures and create pre-aggregated SQL views to load data from the Silver layer into the Gold layer (Noureldien Mohamed Mahmoud)
- Begin scaffolding the Python pipeline script by setting up local database connection modules (Ali Mohamed Ahmed)

Deliverables:

- Operational Silver layer handling data standardization and incremental Merges/Upserts.
- Completed Gold layer featuring a structured Star Schema and executive SQL views.
- Initial framework and repository structure for the Python automation script.

Week 3: Pipeline Automation, Power Query & Data Modeling

- Build and encapsulate the end-to-end Python pipeline to execute all Stored Procedures (Staging to Gold) with 1-click, including TRY/CATCH error handling (Ali Mohamed Ahmed)
- Connect Power BI to the SQL Server Gold views using Import Mode (Momen Shreif Mohamed)

- Apply any necessary final data preparation steps using Power Query inside Power BI (Momen Shreif Mohamed)
- Build the Power BI data model, ensuring relationships guarantee fast and seamless queries (Momen Shreif Mohamed)
- Write and optimize the core DAX measures and calculations required for the analytical visuals (Momen Shreif Mohamed)

Deliverables:

- Fully functional, 1-click Python automation pipeline with error logging.
- Power BI file connected to the Gold layer with an optimized data model.
- Completed Power Query transformations and foundational DAX measures.

Week 4: Dashboard Design, UI/UX & Final Validation

- Translate stakeholder User Stories into role-specific dashboard pages (CEO, CFO, COO, Churn) (Mohamed Gehad Mohamed)
- Design the UI/UX with interactive elements like drill-downs, slicers, and tooltips (Mohamed Gehad Mohamed)
- Write any final, visual-specific DAX adjustments needed for the dashboard elements (Momen Shreif Mohamed)
- Conduct rigorous data validation, comparing dashboard figures to the SQL database to ensure visual accuracy (Mohamed Gehad Mohamed & Momen Shreif Mohamed)
- Format and polish the final dashboard aesthetics for executive presentation (Mohamed Gehad Mohamed)
- Push all final SQL scripts, Python code, and documentation to the GitHub repository (Mostafa Mahmoud Abdelhamid & All Team Members)

Deliverables:

- Completed, interactive Power BI dashboards for all required stakeholders.
- Verified data accuracy report ensuring visual metrics match the database.
- Final project ready for delivery with the code base fully committed and organized on GitHub.

2- KPIs:

1. Pipeline Performance & Reliability

- **Pipeline Execution Time:** The total time it takes for the Python script to run the entire process (extracting data from Supabase/CSV, and running Staging, Bronze, Silver, and Gold transformations).
    - *Target:* Less than 5 minutes (given the 100MB data size).
- **Pipeline Success Rate:** The percentage of automated Python pipeline runs that complete successfully without triggering a SQL TRY...CATCH error.
    - *Target:* 99% success rate.

2. Data Quality & Accuracy

- **Data Accuracy Match:** The percentage of key metrics (like Total Revenue or Churn Rate) in the final Power BI dashboard that exactly match manual SQL queries run against the raw source data.
    - *Target:* 100% accuracy.
- **Data Cleansing Rate:** The percentage of rows that successfully pass from the Bronze layer to the Silver layer without being dropped due to schema drift or unresolvable format errors.
    - *Target:* > 98% of source rows successfully processed.

3. Business Intelligence (BI) & User Experience

1- **Dashboard Refresh Time:** The time it takes for Power BI (Import Mode) to fully load and refresh the data model from the SQL Server Gold layer views.
    - *Target:* Less than 2 minutes.

2- **Stakeholder Adoption Rate:** The frequency with which the end-users (CEO, CFO, COO) log in and interact with the dashboards (using slicers, drill-downs) during the first month after delivery.
    - *Target:* At least 2-3 logins per week per executive.

3- **Requirement Fulfillment:** The percentage of the original business questions (defined in Week 1 User Stories) that are successfully answered by the final dashboard.
    - *Target:* 100% of agreed-upon questions answered.

3- Risk Assessment & Mitigation Plan:

1. Database Loading & Staging Risks

Risk: Bad CSV Data

- **Impact:** Medium. The CSV files have errors (like letters instead of numbers), causing the database to crash when loading.
- **Mitigation:** Load the raw CSV data as plain text first, then use the Silver layer to clean up the text and fix the errors.

2. Transformation & Architecture Risks

Risk: Data Duplicates

- **Impact:** High. The automated merge accidentally copies the same data twice, making the final metrics wrong.
- **Mitigation:** Always use unique ID numbers or a combination of critical IDs (like Customer ID ect …) to check if a row already exists before adding it.

3. Pipeline Automation Risks

Risk: Python Fails Silently

- **Impact:** High. The Python pipeline crashes halfway through without notifying anyone, leaving the database half-finished.
- **Mitigation (Ali):** Use TRY...CATCH in SQL, and make the Python script stop immediately and print the exact error location.

4. BI & Stakeholder Delivery Risks

Risk: Wrong Dashboard Numbers

- **Impact:** Critical. The dashboard shows different numbers than what the database actually has, causing a loss of trust.
- **Mitigation:** Manually write SQL queries to check the totals and compare them directly to the Power BI numbers before presenting and Checking the data model architecture.